Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the test accuracy by dataset.

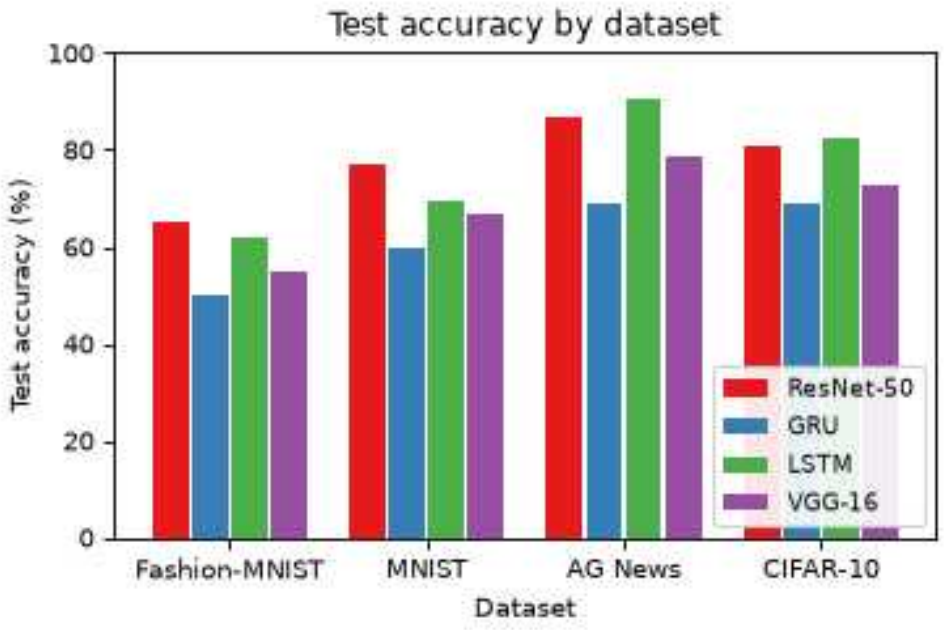


Figure 1: Test accuracy by dataset for the models in this experiment.

On CIFAR-10, VGG-16 obtains a test accuracy of over 70%. ResNet-50 reaches 63.2% test accuracy on Fashion-MNIST.

Figure 2 shows the accuracy during training.

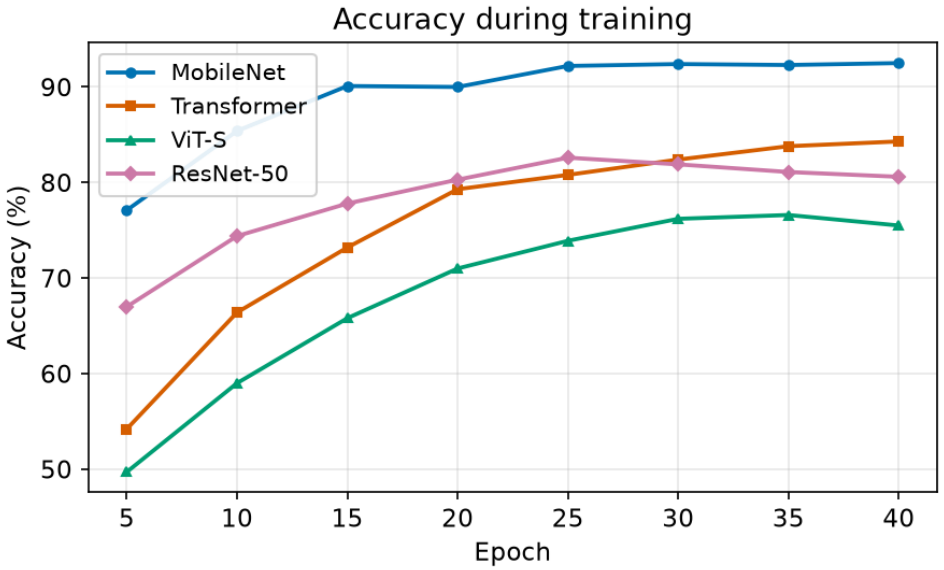


Figure 2: Accuracy during training for the models in this experiment.

As shown in Figure 2, Transformer outperforms ResNet-50 by 26.2 percentage points at epoch 35. As shown in Figure 2, at epoch 20, ViT-S achieves 71.0% accuracy.

Figure 3 shows the accuracy during training.

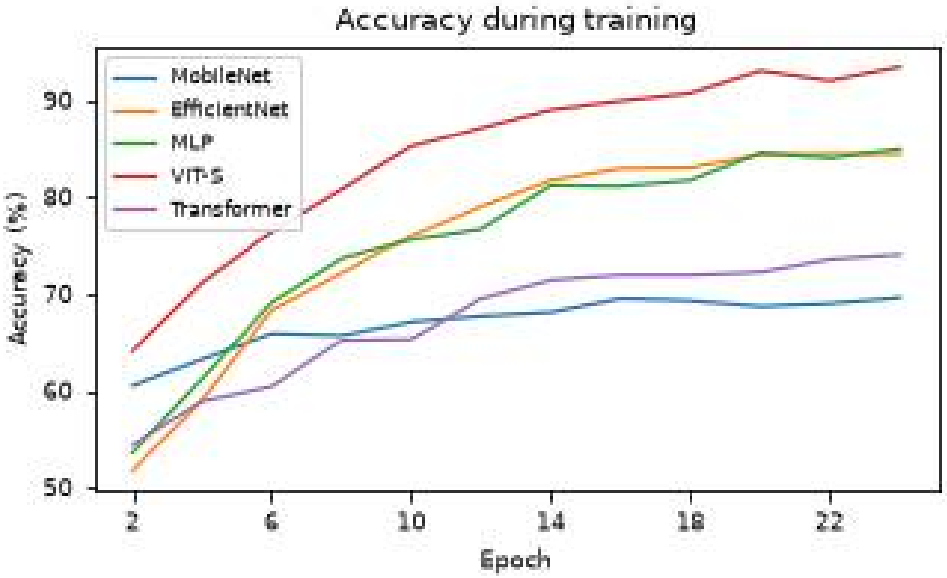


Figure 3: Accuracy during training for the models in this experiment.

As shown in Figure 3, MLP outperforms Transformer by 8.7 percentage points at epoch 6. ViT-S attains a accuracy of about 90% at epoch 12 (Figure 3).

Figure 4 shows the accuracy during training.

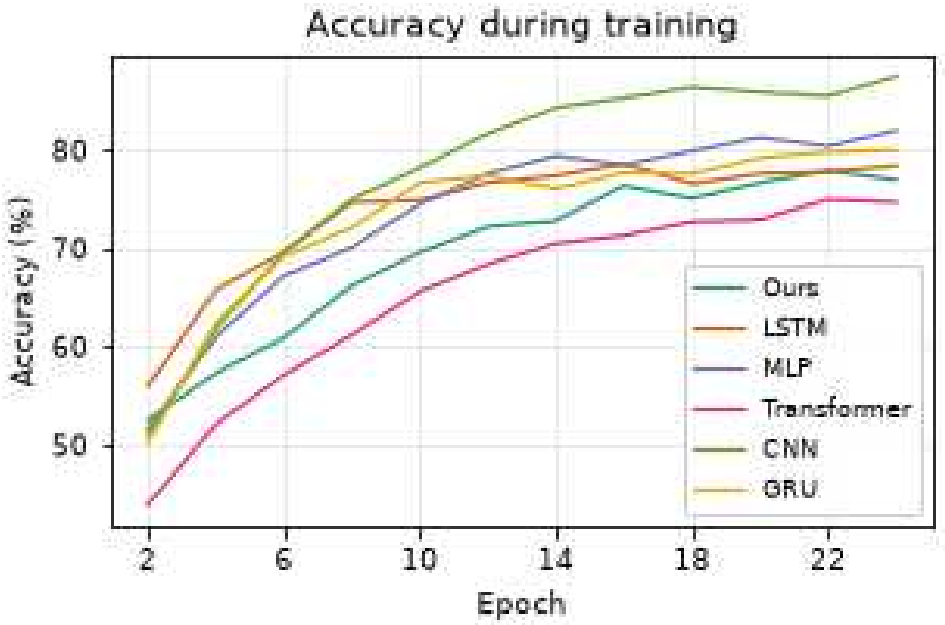


Figure 4: Accuracy during training for the models in this experiment.

At epoch 2, MLP achieves 52% accuracy (Figure 4). As shown in Figure 4, CNN outperforms Ours by 32.9 percentage points at epoch 16. At epoch 10, CNN is 9.6 points better than Ours (Figure 4).

3 Discussion

Logistic Regression reaches 76.7% accuracy on STL-10. We note that the training was stable for all models.